

Nursery Design

Healthy transplants are an important part of the equation for successful harvests, and that is why it is necessary to plan a suitable location for their production. This document summarizes the factors to consider when planning a nursery.

Why Use Transplants and Produce Them Yourself

Although the propagation of certain crops via transplant is not possible, using transplants has several advantages over direct seeding:

- Gives a head start on weeds at planting time
- Allows for precise spacing when planting, maximizing the use of the bed and increasing production
- Reduces the number of seeds needed
- Saves time in the garden
- Crops can be started earlier

Buying transplants from another grower is a common practice in Europe, but the advantages of producing your own transplants are numerous:

- More flexibility and control over cultivar selection, production schedule and transplant quality.
- No delivery problems or delays
- Less risk of importing insects or pests
- Pride and satisfaction

Estimating Space Requirements

The first step when planning a nursery is to calculate your space needs. You should take into consideration the fact that the business may grow and the space required will change. It is better to plan for more space than to find yourself cramped in a few years, considering that building an annex or a second infrastructure is much more expensive and complicated.

To estimate the space requirements, it is necessary to know the growing area needed when the nursery is at its busiest, generally just before the outdoor conditions become favorable for field planting. This calculation can be made using a production schedule, by adding up the surface area required for each growing container, bearing in mind that pots may need more space for aeration and optimal development.

To this must be added the space required for any other activities that we would like this area to be used for, for example:

- Workbenches
- Space to circulate (depending on the tables, see below)
- Storage space (trays, soil, pots, etc.)
- Tool storage

Positioning and Orientation of the Nursery

The nursery should be located in a central area. The plants in it are fragile and need frequent care.

Soil quality is not an issue, unless you plan to grow crops in the ground. A well-drained, flat, sunny and wind-sheltered location is ideal. The installation of a windbreak hedge helps to reduce wind speed, but it must be positioned far enough from the greenhouse so that it does not block the light (at any time of the day, all year round) and to avoid branches damaging the structure or the covering.

It can be useful to orient the greenhouse in the same direction as the prevailing winds. This is especially true in a northern climate. This way, snow accumulation on the greenhouse structure can be reduced, maximizing light transmission and reducing the load on the structure.

Insulating the Perimeter of the Greenhouse

The busiest time of the year in the nursery is during the production of transplants at the beginning of the season. Since this is a time when it is often still cold outside, insulating the perimeter of the greenhouse is an attractive option that will result in reduced heating costs. Also, since the trays will be placed on tables, you can afford to raise the insulation without cutting access to light.

Completely insulating or using an existing building for the north wall can also help to save energy.

A simple and effective way to optimize heating is to ensure that leaks and air infiltration at openings, holes, joints and cracks in the greenhouse structure are sealed.

Automated Climate Management System

A climate management system is essential to ensure and maintain ideal climatic conditions for plant development. As in a greenhouse, it consists of a sensor and a controller that will be connected to the heating and ventilation systems. In theory, the climate of a nursery is managed just like the climate of a greenhouse, but the nature of the climatic challenges is different. Since there is a large amount of plant surface in a greenhouse, evapotranspiration is significant and it is often difficult to bring down the humidity. In a nursery, as there is a small volume of plants that transpire, it is rather the opposite. We try to conserve moisture and keeping it high enough can be a challenge. It also makes it difficult to lower the temperature on hot, sunny days.

An automated system is used to pre-program set points for climatic parameters (temperature, humidity). The sensor collects data about ambient conditions and the controller turns the heating and ventilating equipment on and off until climatic conditions are within the set points.

It is important to place the sensor in a place that is representative of average climatic conditions in the nursery. If the sensor is placed in a particularly cold place (near a door or in a draft for example), the system may generate too much heat. If it is in a particularly hot place (too close to a heating source or high up), the system will keep the temperature of the greenhouse too low. In order to collect useful data, the sensor must be placed in a location that is representative of the average climate conditions as perceived by the plants, usually in the center of the greenhouse, about 12-16 po (30-40 cm) above the trays.

Heating System

The most important quality of a nursery heating system is its reliability. Unit heaters should be chosen for their ease of maintenance, ease of repair and price. A newer system and new heaters will also be more energy efficient.

Determining heating needs

To effectively plan the heating system, you will need to anticipate the BTU requirements in the worst case scenario, at peak demand. Heating requirements vary depending on:

- The volume of air to be heated
- The maximum difference between the desired indoor temperature and the outdoor temperature
- The type of covering

BTU requirements calculators for greenhouse heating are available online.

Energy source

Although there are other energy sources, propane remains a simple and cost-effective solution. Depending on the result of the BTU requirement calculation, it is advisable to split the capacity over 2 smaller heating units, for safety.

Heat distribution

Since warm air tends to rise, circulating heat under the tables at floor level is an effective way to heat the nursery's air volume. This is achieved by connecting the heaters to perforated polyethylene heating tubes through a duct.

HAF fans are also used to even out the climatic conditions. See the section on ventilation systems.

Backup system

Each furnace should also have its own thermostat, programmed a few degrees below the controller's set point. In the event of a malfunction or error, it will be activated and the worst will be avoided.

It is also advisable to equip yourself with an alarm system that allows you to send alerts to a phone in case temperatures rise or fall too much.

Power failure

Without any electricity, the controller, the temperature sensor, the furnace blower and the vents will not work. Combined with the alarm system, an emergency power supply (gasoline generator, automatic or not) can save the season, especially in northern climates.

Ventilation System

While the cold may be a problem in winter, it is not an issue in summer. Creating heat through fuel combustion is technically easier than creating cold in the summer.

The ventilation system is used to cool the nursery down when the temperatures are too high. Doing so maintains the ideal conditions necessary for the health of the seedlings.

Roll-up sides

The roll-up sides are an important part of the nursery's passive ventilation system. When properly programmed, they will gradually open and close based on the difference between the indoor temperature observed by the sensor and the set point.

Open roof and/or positive pressure ventilation

A positive pressure ventilation system or an open roof can complement the work of the roll-up sides.

An open roof greenhouse allows you to passively evacuate the heat accumulated overhead inside the structure. This ventilation equipment is very efficient, but it is a choice that will increase the total cost of the structure and make its construction more complex. It may be a useful strategy for a nursery that has a large surface area, because the bigger the greenhouse is, the less efficient roll-up sides are at ventilating.

The positive pressure system is a forced air ventilation system. It consists of a fan that is installed at the top of one of the end-walls of the greenhouse and a large perforated tube that extends through the roof of the greenhouse from one end to the other. When the temperature in the nursery rises above the set point, the fan turns on and pushes the outside (cooler) air inside. Because it is heavier, the cooler air falls down on the crops and the warm air that is there is displaced and evacuated through the opening sides. A positive pressure system is less expensive and less complicated to install than an open roof system. It also has the advantage that it can be installed at a later stage, after the greenhouse is built.

HAF

HAF (Horizontal Air Flow) fans are used to circulate air in the nursery. This strategy reduces the risk of fungal diseases, mixes the air layers, evens out the climatic conditions (temperature, humidity and CO₂) and promotes the production of stocky and robust transplants. They operate continuously and can reduce heating costs by up to 30%.

The ventilation capacity required is equal to 2 to 4 times the greenhouse floor area in square feet. For example, the ventilation capacity required in a 30' x 100' greenhouse is estimated at between 6000 cfm (30' x 100' *2) and 12000 cfm (30' x 100' *4). It is therefore necessary to equip yourself with HAFs that can move this volume of air (expressed in CFM (Cubic feet per minute)). To promote consistent climatic conditions, it is better to have a greater number of smaller fans instead of just a few powerful ones.

When installing, fans should not be closer than 10-15 feet (3-4.5m) from the nearest wall. The space between HAFs should not be more than 50-60 feet (15-18 meters), or 30 times the diameter of your fan. They should be securely fastened to prevent movement during operation. Fans should be installed under the thermal screen, if you have one, and about halfway up the greenhouse, or at gutter height. Fans should be placed in a way that forces air to circulate in a circular pattern. Air movement should be continuous to avoid the accumulation of stagnant air.

Access to Water

For crop health, it is important to have good access to quality water.

Irrigation water should be tested for quality. The quality of water for seedling production depends on:

- Its temperature
- Its pH
- Its mineral content
- Its microorganism content

Cold water is stressful to crops and depending on its source, a mitigation strategy may be necessary. A plastic tank can be installed in the nursery to store the water until it reaches an appropriate temperature. A transparent tank encourages algae growth, unlike an opaque, black tank. A pump and a pressure tank will be needed to ensure irrigation from this water reserve.

Note that for microgreens production, the Canadian Organic Standard prescribes the use of potable water for irrigation.

Depending on the production schedule, the objectives and the geographical context of the business, a frost protection valve may be required.

Although it is possible to automate the irrigation of trays and pots, either with a sprinkler or a spaghetti system, the time-saving potential seldom justifies it, especially for diversified production. Crops in the nursery are not necessarily in the same types of trays, pots, potting soil and they are at different stages of growth. Their water needs are therefore different and watering should be adjusted accordingly.

In addition, hand irrigation forces regular observation of the plants and monitoring of the crops. For pleasant and efficient manual irrigation, hoses should be of sufficient length to reach all trays. A hose reel or hanging system will keep these hoses stored away when not in use.

Misting

Keeping the relative humidity around 80% in the nursery can be a challenge. The small leaf area of the plants leads to little evapotranspiration, unlike in greenhouses where mature plants are growing.

As a result, a strategy may be needed to add water to the air and increase the relative humidity. The installation of a misting system could be a useful solution to this.

The misting process consists of projecting water in fine droplets, thus creating a mist. You can program various cycles according to a schedule during dry weather, or have the system start automatically according to the reading of the humidity sensor.

The misting system is also an effective mechanical control for spider mites, which thrive in hot, dry weather.

The misting system can also be part of the temperature management strategy in very hot weather. The fine water droplets that settle on the leaves evaporate, taking some of the heat from the plants with them.

Floors

There are several options to consider when choosing a type of floor for your nursery. Here are some characteristics of the most common options.

Dirt floor

- Cost-effective
- Uneven
- Difficult to move rolling equipment
- Weed management necessary
- Mud accumulation possible
- Impossible to sweeping
- Can be recultivated

Dirt floor, covered with nursery landscape fabric (white or black woven ground cover)

- Woven ground cover must be treated to be UV resistant
- Rolling equipment can be moved around, but not ideal
- Little to no weed management
- Easy to sweep

- Can be recultivated
- Cost-effective

Concrete slabs

- Easy to move most equipment on wheels
- Little to no weed management
- Easy to sweep
- Can be recultivated
- More expensive than a dirt floor but less expensive than a concrete floor

Concrete floor

- More expensive
- Permanent
- Ideal for equipment on wheels
- Thermal mass (heat conservation)

The choice of floor type depends on production goals and available budget.

Tables

Trays and pots are placed on tables for the duration of their stay in the nursery. When selecting tables, the following factors should be considered:

- All trays should be easily accessible for maintenance, moving and irrigation.
- Tables should have a length and width that is a multiple of the dimensions of the trays, to avoid ending up with unusable space.
- They should be made of material that is resistant (rust, rot, sunlight).
- The material chosen for the table top must allow water to pass through.
- Specialized equipment is available to maximize production space (see rolling tables)

Homemade tables

Many growers build their own nursery tables from metal mesh, staples, wood, cement blocks and/or old pallets. Their life span depends on the materials used and the quality of the construction. The main advantage is the cost, while the main disadvantage is that they cannot be moved easily and their lifespan can be short.

Rolling tables

In this system the table legs are fixed to the ground and only the top is mobile.

Rolling tables save a lot of space and maximize the area available for crops. All trays must be accessible for watering, but they do not need to be accessible at the same time. With fixed tables, there must be space to circulate around each table or around every two tables, which is actually a lot of space. With rolling tables, a space is created to circulate between 2 tables, leaving all other tables stuck together. To access the next table, simply roll the first table in order to simultaneously close the first corridor and create an opening between the next two tables.

This minimizes the proportion of circulation space in relation to the production area, which is definitely an advantage when you consider the costs of purchasing, installing and heating the nursery.

Storage Space

In general, the nursery's design provides for work and storage spaces. You must first define your needs (and future needs) to properly conceptualize the space and allow for enough room. In addition to growing tables, the nursery can be used to:

- Store trays, pots, domes
- Store seeders, grafting equipment, heat mats, artificial light, extra fans
- Store potting soils, amendments, fertilization (greenhouse and field)
- Have a room temperature water tank
- Store various equipment to protect them from potential frost (tarps, nets, machinery, irrigation)

Overestimating the space you need decreases your chances of having to deal with a space that is too small after only a few years. Space will also allow you to increase efficiency and give you more options and maneuverability.

Work Space

To facilitate nursery work, it is necessary to provide ergonomic and comfortable spaces that can be used for:

- Sowing trays
- Potting up
- Making potting soil (if homemade)

A smooth, solid, stable, easy-to-clean work table is ideal. Ledges on the sides at the back will help contain the potting soil on the workspace. These tables can be fixed, collapsible, or on wheels, depending on your needs. It should have enough height to slide a container on wheels underneath. To promote

ergonomics, the height of the work surface should be at elbow height. Ideally, this workstation should be close to trays, potting soils, seeders and other tools and equipment used for seeding.

Hardening Off Area

All crops, except those that will be planted in heated greenhouses, must be hardened off one week before transplanting. This space may or may not be connected to the nursery. Either way, it is necessary to plan a space where the climate can be managed separately and where the plants can be gradually subjected to outside climatic conditions (direct natural light, temperature variations, wind).

A hardening off area that is connected to the nursery will reduce the time required to move trays, but its design can be more complex. This choice depends on the volume of trays to be transported. If the hardening off space is not adjacent to the nursery, it should be located nearby.

Either way, it is necessary to ensure that water is accessible to irrigate the plants as they harden off. An easy access for the transport of trays to the gardens is also very practical.

Thermal Screen

A thermal screen is a slightly opaque fabric that is mechanically deployed over the crops. It has several advantages:

- Heat retention in cold weather
- Decrease in the volume of air to be heated during the night
- Shading during the summer

A thermal screen can allow savings of 20% to 40% in heating costs. The payback period for this equipment depends, among other things, on the length of the nursery season and the type of infrastructure. An economic study is necessary to determine if this is a worthwhile investment for you.

Other Considerations

Covering

The greenhouse covering that offers the best light transmission and insulation for its price is a 6 mil double polyethylene with air blown between the two layers. You should choose a UV resistant treatment for the outer layer and a

condensation resistant treatment for the inner layer. If you do the installation yourself, make sure to put the plastics the right way round.

Artificial light

Light is sufficient for normal plant growth as soon as daytime is over 10 hours (depending on the latitude - e.g. February 6th in Montreal).

If nursery production is planned before this date, it may be worth considering the use of artificial light to ensure that the plants' needs are met. For a smaller production volume, the indoor germination room is a good option.

Grafting space

Grafting is a delicate operation and success depends almost entirely on the conditions in the healing chamber. The first hours and days are critical and conditions must be precisely managed there for a week. For these reasons, the nursery is not the ideal place for grafting, although it can be done successfully using another structure inside it.

Seed storage

Conditions in the nursery are not ideal for seed storage. It is best to choose a cool, dark, dry place and to put the bags in airtight containers to protect them from rodents.

Insect netting

Insect netting can be useful in the nursery depending on the historic levels of pressure from insect pests. It does not, however, guarantee their absence.

Doors

Choose high quality, insulated doors that close tightly and are large enough to accommodate all equipment and materials.

Case Study: End-of-Greenhouse Nursery

It is possible to use part of a greenhouse for transplant production. A balance will have to be struck between the ideal climatic conditions for both projects, but many growers opt for this solution.

This nursery within a production greenhouse, in whole or in part, can be temporary (only at the beginning of the season) or permanent. In the case of a temporary nursery, the nursery tables are placed on the production beds.

When they are no longer needed, they are simply removed and the space freed up can be cultivated normally. The use of space must be well planned and consistent with the production schedule.

Case Study: Making Sections in the Nursery (Moving the Sensor)

Typically, the season starts slowly in the nursery, with early tomato and onion seedlings. At this time, a very small proportion of the space is used. In order to save on energy consumption, it is possible to use a polyethylene curtain to reduce the volume of space to be heated. In this case, you must:

- Ensure that the division is as tight as possible, without impeding the movement of the opening roof (if applicable) or the thermal screen (if applicable).
- Ensure that the section that will be heated is on the same side as the furnaces.
- Move the sensor so it is located at the center of this new space.
- Limit the heating tubes to the section being heated.
- Ensure that the positive pressure system goes into the new section (if applicable).

Case Study: Using the Nursery for Drying/Curing

Toward the end of the summer, the nursery empties. This is also the time when the fall crops are harvested. Some of them need to be dried to promote good preservation, such as:

- Storage onions
- Shallots
- Garlic
- Ground cherry
- Winter squash (not all types)

With a few modifications, the nursery can become the ideal place for drying:

- Make sure fan speed is at maximum.
- Add fans under the tables. There should be plenty of constant air flow.
- Deploy the thermal screen (if available) to avoid burning drying crops.
- Stop misting (if applicable)

If transplant trays are still in the nursery (e.g. for winter production), avoid harming them and try to balance the optimal conditions for both projects. This includes planning the use of space using the production schedule. The

introduction of insect pests with field crops is also possible. Onions can bring thrips or leek moths into the nursery, for example.

Conclusion

As you can see, planning a nursery is quite complex. Don't hesitate to consult an expert advisor to help you make decisions that are consistent with your situation and goals.